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SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - NOV'2024

B.E – COMPUTER SCIENCE AND ENGINEERING

20CS215 - CLOUD COMPUTING

Answer ALL questions

Duration: 3 Hours

Maximum: 100 Marks

PART - A

(7x 1 = 7 Marks)

1. **CO1** What is a primary advantage of distributed systems?
 - a) Centralized control
 - b) Reduced latency
 - c) Limited scalability
 - d) Simplified maintenance

2. **CO1** What advantage do virtual clusters offer compared to traditional physical clusters?
 - a) Lower security risks
 - b) Reduced scalability
 - c) Higher hardware costs
 - d) Increased flexibility and agility

3. **CO2** Which security measure helps prevent unauthorized VM cloning?
 - a) Network segmentation
 - b) Data encryption
 - c) Secure boot
 - d) Patch management

4. **CO3** How does multi-factor authentication (MFA) contribute to identity and access management?
 - a) By requiring users to provide multiple forms of authentication before granting access
 - b) By encrypting data during transmission
 - c) By monitoring network traffic for suspicious activity
 - d) By isolating virtual machines from each other

5. **CO4** How does Docker differ from virtual machines (VMs)?
 - a) Docker containers share the host operating system kernel, while VMs have their own kernel
 - b) Docker containers are slower to start compared to VMs
 - c) Docker containers are larger in size compared to VMs
 - d) Docker containers require a hypervisor for isolation, while VMs do not

6. **CO3** Which of the following services in AWS is used for automating code deployment to any instance, including Amazon EC2 instances and instances running on-premises _____.
 - a) AWS Elastic Beanstalk b) Amazon RDS
 - c) AWS Lambda d) Amazon SNS

7. **CO4** Which of the following is NOT a command for managing Docker containers _____.
 - a) docker run b) docker start c) docker stop d) docker build

Fill in the blanks:

(3x 1 = 3 Marks)

8. **CO1** In virtual clusters, a _____ is a logical representation of the resources assigned to a specific project or team, including virtual machines, storage, and networking resources.

9. **CO3** The _____ in Kubernetes is responsible for scheduling and running containers on nodes in the cluster, ensuring that the desired state of the system is maintained.

10. **CO4** _____ is a cloud computing service model where cloud resources are shared among multiple organizations, and customer's access and manage their own resources through a web browser or API.

PART - B

(5x 3 = 15 Marks)

11. **CO1** Write some XaaS significance in modern IT infrastructure.

12. **CO1** Compare and contrast the key features of Platform as a Service (PaaS) and Software as a Service (SaaS) layers in cloud computing architecture.

13. **CO2** How do identity providers contribute to authentication in cloud computing?

14. **CO3** How does Docker Hub facilitate the sharing and distribution of container images?

15. **CO4** Outline the role of the Kubernetes API server and how it interacts with other components.

PART - C

(5 x 15 = 75 Marks)

Q.No: 16 Compulsory Question

Q.No: 17 to 22 Answer any FOUR Questions

Compulsory Question:

16. **CO1** i) Discuss the evolution of cloud computing, highlighting its key milestones, technological advancements, and the impact it has had on modern computing paradigms. (8)
- CO1** ii) Explain the layers of cloud computing architecture, and how do they interact with each other? (7)
17. **CO1** i) Discuss the different deployment models in cloud computing, highlighting their key characteristics and advantages. (8)
- CO1** ii) Describe the role of physical infrastructure in data center architecture (7)
18. **CO1** Discuss the various virtualization structures, tools, and mechanisms used in modern IT environments. Provide examples of each and explain their significance in improving resource utilization, scalability, and manageability. (15)
19. **CO1** i) Your company, an e-commerce retailer, is looking to migrate its on-premises infrastructure to the cloud to improve scalability and reduce operational costs. Outline a comprehensive cloud migration strategy using AWS services (8)
- CO1** ii) Describe the benefits of using virtual memory, including increased address space and efficient memory management. (7)
20. **CO1** i) Discuss the various security risks associated with the sharing of images on social media platforms. (8)
- CO1** ii) Examine the evolving landscape of data privacy and security issues and also provide the main challenges faced by individuals, businesses, and governments in safeguarding sensitive data. (7)

21. **CO1** i) Explain the significance of Microsoft Azure in the cloud computing. (8)
Discuss the key features and services offered by Azure
- CO1** ii) Discuss the importance of authentication in cloud computing (7)
environments. Explain the role of authentication in ensuring secure
access to cloud resources and protecting against unauthorized access and
data breaches.
22. **CO1** i) Discuss the data management in Docker container environments. Analyze (8)
the challenges and considerations involved in handling data within
Docker containers, including persistence, portability, and security.
- CO1** ii) Explain the deploying Kubernetes in Docker (KinD) environments. (7)
